Geography of X* Energy, Vitality, and Happiness Across US Counties

Saturday 8th August, 2026 10:03

We use X (Twitter) data to study Energy, vitality, and happiness Across US Counties. Energy, vitality, and happiness is measured based on human expression on X. Multiple metrics are used. One contribution is geography of energy/vitality: TODO briefly explain like in 3 sentences using some descriptions and maybe even tweet examples. Another part of the paper is measuring Subjective Wellbeing (SWB). As found earlier in the literature cities tend to be less satisfied with life (Okulicz-Kozaryn and Valente 2021), but there are important nuances: TODO quaadratics etc

1 Introduction

Terms SWB, happiness, and life satisfaction are usually used interchangeably in happiness and social indicators literatures. But specifically they mean rather cognitive global subjective self-evaluation of one's life as a whole rather than momentary affective happiness. Here it rather the latter, momentary affective happiness as human expression is recrded via tweets on X.

Reliability and validity of twitter data discussion is postponed to section "Data."

Much ink has been spilled on geography of Subjective Wellbeing (SWB), including in the US, in particular there are studies across counties (e.g., Okulicz-Kozaryn and Mazelis 2018, Winters et al. 2015, Okulicz-Kozaryn et al. 2024), and across urban rural continuum (Okulicz-Kozaryn 2024, 2023, Okulicz-Kozaryn and Valente 2020). Major findings have been that humans are less satisfied with their lives in populated dense North East, but also in poorer Midwest, and some of the lowest life satisfaction scores in addition to largest cities such as NYC and Philadelhia, have been reorded in South, in Particular in poor Alabama, Mississipi, and Louisiana. Greatest life satisfaction, on the other hand, was in sparsely populated North and West, in states such at Nebraska and Utah, but not in overpopulated California.

That has been in SWB and geographic literatures. One limitation of these literatures is sole use of survey data such as GSS and BRFSS and lack of energy/vitality metrics. One contribution of this paper is to connect these literatures with X literature, which seems to be mostly in computer science.

2 Twitter geography of happiness literature

There is separate X (Twitter) literature, starting with pioneering widely cited paper on overall geography of twitter: Leetaru et al. (2013), but it does not study energy/vitality nor happiness. Various subsecquent studies did look at boradly defined energy/vitality and happiness, but they differ in important ways from our study.[1]

There is an intersting and related doctoral dissertation (Gupta 2018), but analyzes geographies in UK. Like us, Jaidka et al. (2021) is an analysis of US counties and urban-rural, but in terms of stress.

*Twitter

[1] Except another pioneering study, Mitchell et al. (2013), that we discuss towards the end of this section.

Chai et al. (2023) produced "twitter Sentiment Geographical Index (tSGI), a location-specific expressed sentiment database with SWB implications", but it doesnt really analyze energy/vitality nor happiness and has a global, not the US, focus.

There are neighborhood level studies in NYC studying happiness and poverty (Tan and Guan 2021) and proximity to parks (Plunz et al. 2019). And one in Detroit MI analyzing happiness and neighborhood conditions (Park et al. 2021). And another analyzing happy tweets, healthy food references, and physical activity across neigborhoods in Salt Lake, San Francisco and New York (Nguyen et al. 2016). Anothother study analyzes sentiment across spatiotempoprtal dimensions (land use and times of the day) across Massachusetts (Cao et al. 2018).

Then there are several studies that are closer to ours. Hollander and Renski (2022) studies cities, like us, but excludes rural areas unlike us, and only analyzes 50 declining cities matched with 50 growing/stable cities and only using 300k tweets.

Even though Hu et al. (2019) may seem similar to our study as it bradly asses sentiment and positive emotions across us counties, it is quite different. Their approach and framing are very different. Rather than focusing on geography of wellbeing like us, they explore tweeter topics and correlations in space. Rather than energy/vitality or happiness, they study variuos topics such as birthdays, parties, and great time on sunday and focus on spatial correlations rather than levels like us.

The closest to what we do is Mitchell et al. (2013) like us focus on geography within the US, but they study states (we study counties), and they study also some cities like us, but only a limited set, while our data covers virtually all of united states including rural counties. They only have 80m words generated in 2011 on X, but we use a much larger dataset of over 1b tweets 2010-2019. While they also show some results for all of the US like us in their map in figure 5, it is difficult to read for less populated areas. Another limitation that we improve on is the use of multiple energy/vitality and happiness metrics as opposed to just one happiness metric in their study. Also note the Atlantic article that discuss the arxiv version of their study.[2] We discuss Mitchell et al. (2013) more as we present our results. Again, note that Mitchell et al. (2013) is one of the pioneering studies done over 10 years ago.

Last, but not least there is an arxiv preprint (Iacus and Porro 2025) by producers of the data. We both study US geography of happiness, and we both use life satisfaction and happiness metrics. But here similarities end. In addition to life satisfaction and happiness metrics we use 7 other measures. Iacus and Porro (2025) measure rural-urban with USDA ERS 9 Rural-Urban Continuum Codes; we use population deciles. Iacus and Porro (2025) doesnt analyze the data across US counties, only discusses extreme cases–23 most rural counties in their table 4. Finally, Iacus and Porro (2025) in addition to geography also focus on politics and income, while we focus just on geography and dig deeper. Thus, despite using 2 variables from the very same dataset, our studies differ in critical ways.

Curiously Kjell et al. (2023) shows a map simialr to our county estimates in fig 5.1C and cites Jaidka et al. (2020). Jaidka et al. (2020) focuses on comparison of twitter happiness with Gallup data but doesn't analyze by urban-rual and county-level estimates cited in WHR are not there, not even in supplemtary material, hence it is difficult to discuss. But there are differences: we have multiple measures, do analysis by urban rural. And, crucially, the results that overlap, county level happiness maps, the findings differ. We discuss more when discussing our maps.

Overall, while there is twitter geography of happiness literature, it either focus on different geographies (other countries, or other aggregations such as state or neghborhoods) and different (e.g., stress) or more limited metrics such as happiness only.

3 Data

Data are from `https://dataverse.harvard.edu/file.xhtml?fileId=13094135&version=3.1` using file `flourishingCountyYear.csv`
Description of the data is at `https://arxiv.org/pdf/2511.03915` (Iacus et al. 2025).

Here we introduce the Human Flourishing Geographic Index (HFGI), derived from analyzing approximately 2.6 billion

[2]https://www.theatlantic.com/technology/archive/2013/02/the-geography-of-happiness-according-to-10-million-tweets/273286/

geolocated U.S. tweets (2013-2023) using fine-tuned large language models to classify expressions across 48 indicators aligned with Harvard's Global Flourishing Study[...].(Iacus et al. 2025, p. 1)

Note that while data description says 2013-2023, in the actual dataset we also found observations 2010-2012, but it was only 7.9% of the whole dataset, out of which <1% before 2012.

We cut out 2020-2023 due to Covid19 pandemic, but that still leaves us with over 1b tweets. Pandemic affected US geographies very differently and thus would be likely to bias our results–low wellbeing would have been attributed to place, while it was due to Covid19.

We also dropped county-year-variable cells with fewer than 5 valid tweets. If there were very few tweets, the average could have been often more extreme than average from larger sample and thus bias disproportionatelly 2013-2019 averages

Normally one could frown upon Twitter data, but here there are actually reasons to be optimistic. There are $>$ 10 wellbeing measures, and data covers 2012-2019, having it collapsed for each county over 8 years makes sample size bigger and provides more reliability.

¿¿¿here these artcles on reliability etc

We dont get here into twitter data processing and technicalities of natural language processing–it is not our area and we didnt do this–again, we just downloaded pre-processed data and the documentation is in Iacus et al. (2025). We want to focus here on the meaning of the variables instead.

TODO right do have these for all key vars here and comment out in app

Life Satisfaction. Satisfaction with one's life as a whole (`lifesat`).

low: "My living situation is exhausting and unfair."

med: "Things are slowly falling into place–can't complain too much."

high: "Watching the sunset from my porch tonight, feeling truly content."

it is har to believe for us urban america, that there exists rurla america (Hanson 2015), and that in fact there are about 300 counties (yes whole counties not just villages) with pop $<$ 5k!

4 Results

4.1 Urban-rural gradient

population 2015 decile	eneI	posfunct	vitality	energy
88-5.1k	0.35	0.37	0.26	0.43
5.1k-8.9k	0.35	0.35	0.26	0.43
8.9k-13.5k	0.34	0.33	0.25	0.43
13.5k-18.5k	0.33	0.32	0.24	0.43
18.5k-25.6k	0.32	0.31	0.22	0.42
25.6k-36.7k	0.32	0.31	0.21	0.43
36.7k-52.9k	0.31	0.30	0.20	0.42
52.9k-92.6k	0.31	0.30	0.20	0.42
92.6k-204.9k	0.31	0.29	0.20	0.42
204.9k-10077.3k	0.31	0.30	0.21	0.42

Figure 1: Note we stick to `lifesat` tag (life satisfaction) as in the origan dataset, but again it is rather momentary affective happiness than global cognitive evaluationn of one's life as a whole, life satisfaction term used in SWB literature.

population 2015 decile	swbI	lifesat	happiness	innerpeace	optimism	purpose	hope
88-5.1k	0.24	0.15	0.24	0.13	0.15	0.38	0.40
5.1k-8.9k	0.27	0.14	0.27	0.16	0.19	0.41	0.43
8.9k-13.5k	0.27	0.12	0.29	0.16	0.21	0.42	0.44
13.5k-18.5k	0.27	0.11	0.29	0.16	0.21	0.40	0.43
18.5k-25.6k	0.26	0.10	0.29	0.15	0.21	0.39	0.43
25.6k-36.7k	0.26	0.09	0.30	0.15	0.22	0.39	0.44
36.7k-52.9k	0.25	0.08	0.29	0.14	0.20	0.37	0.42
52.9k-92.6k	0.25	0.08	0.29	0.14	0.20	0.37	0.43
92.6k-204.9k	0.24	0.07	0.28	0.13	0.20	0.36	0.42
204.9k-10077.3k	0.24	0.06	0.28	0.13	0.19	0.35	0.40

Figure 2: This figure was pulled directly from a web URL.

4.2 Across counties

It is useful to compare our results here to Mitchell et al. (2013) and discuss the Atlantic article that discuss the arxiv version of their study.[3]: TODO

Kjell et al. (2023) shows a map simialr to our county estimates in figure 5.1C citing Jaidka et al. (2020). It looks like their sample of 1.5b is close to ours. But the results are different possibly due to their "interpolation of missing counties provided through a Gaussian process model using demographic and socioeconomic similarity between counties" (Kjell et al. 2023, fig. 5.1C caption). Alabama, Louisiana, and Missisipi are not unhappy, neither is North East, but Midwest is least happy (we find it unhappy but not as unhappy as Alabama, Louisiana, and Missisipi.[4]

Note the maps with detailed picture more color bins are in appendix. Overall they are very similar and simplified exposition is presented here but the more important differences are:

Results in figure 4 broadly agree at state level with "Figure 1. Average word happiness for geotagged tweets in all US states collected during calendar year 2011" in **?**. Much of the results in our figure 4 also seem to agree at county level with "Figure 5. Map showing happiness of all tweets collected from the lower 48 US states during 2011" in **?**, but it is difficult to say as map averages at county level, while they map closer to actual tweet locations and there are multiple points per county and it is difficult to estimate the averge.

4.3 Zooming in on largest cities

again figure 5.1C with city names all appear to be the very happiest counties of all (dark green).

while the black dot covers most or all of the city-counties, but even the metropolitan area bordering counties that are visible and very happy, which already is surprising

Except Los Angeles and Miami (2nd shade of green, still very happy), and Phoenix (yellow; avergage happiness), and per New York and New Orelans cannot really tell as the black city dots cover polygons. Likewise difficult to tell per other cities due to black big dots, but it seems they are from West to East: Seattle, Los Angeles, Salt Lake City, Pheonix, Denver, Minneapolis, Chicago, Houston, Atlanta, Washington, and Miami

It is important to keep in mind that all data including population pertains to a county, not city, for instance Santa Ana is only about 300k but Orange county is about 3m. Still using cities for ease of iterpretations.

[3]https://www.theatlantic.com/technology/archive/2013/02/the-geography-of-happiness-according-to-10-million-tweets/273286/

[4]But maybe the most striking difference is that urban counties marked in their figure 5.1C with city names all appear to be the very happiest counties of all (dark green). Except Los Angeles and Miami (2nd shade of green, still very happy), Phoenix (yellow; avergage happiness), and per New York and New Orelans cannot really tell as the black city dots cover polygons. Curiously, World Happiness Report and Gallup argue urban happiness despite data showing the contrary–see discussion in Okulicz-Kozaryn and Valente (2021). For instance see `https://news.gallup.com/opinion/gallup/315857/degree-urbanisation-effect-happiness.aspx`.

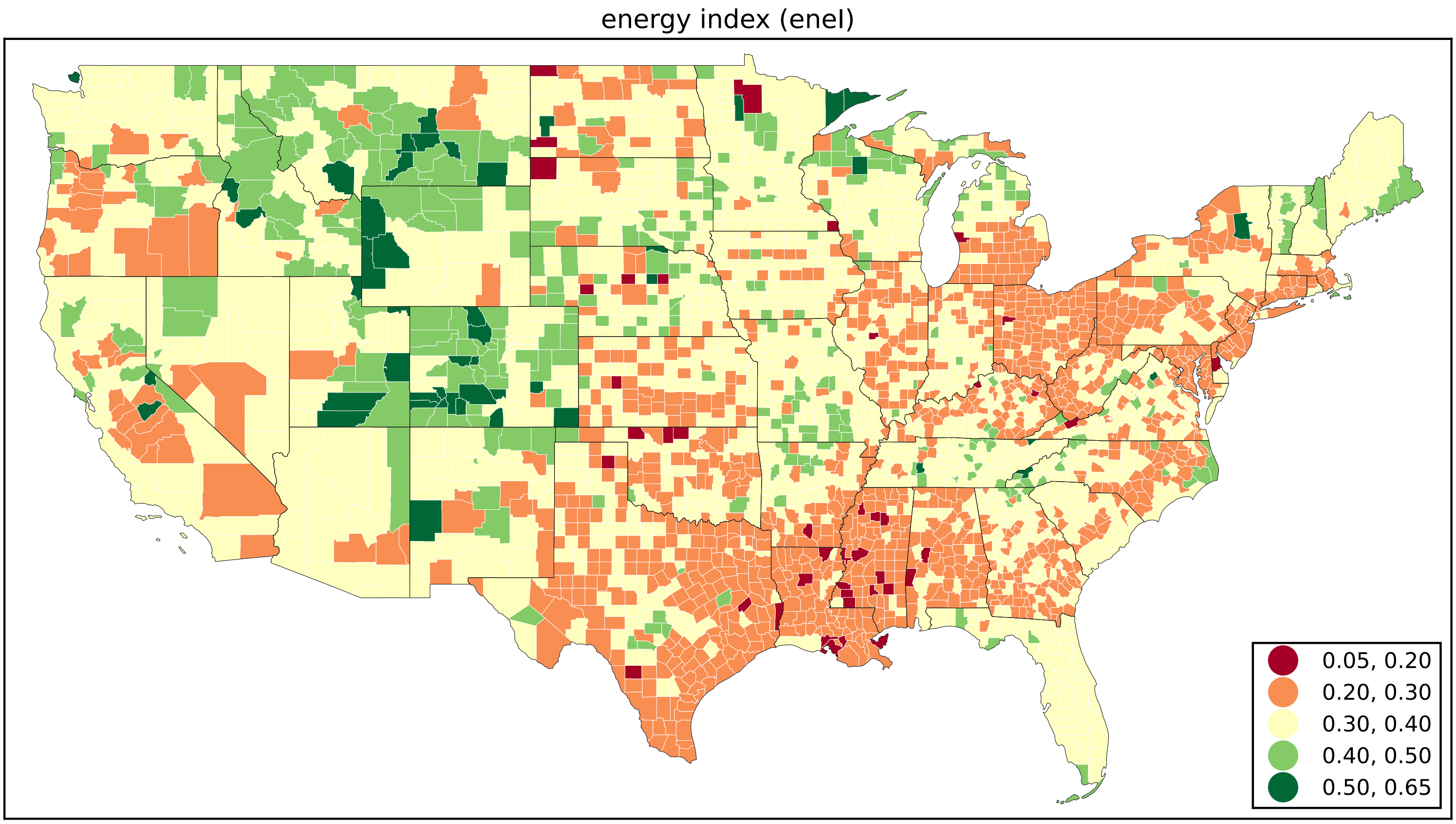

Figure 3: This figure was pulled directly from a web URL.

First,for ease of interpretation summarizing all indicators as happy or unhappy as for most counties there is a clear pattern.

First 4 numeric columns are energy/vitality (enel, posfunct, vitality, energy) and subsequent 6 numeric columns are happiness (swbl lifesat happiness innerpeace optimism purpose).

Happy counties are sunny places, but outside of unhappy Texas, and many coastal like Los Angeles, Phoenix, San Diego, Santa Ana, Miami, Las Vegas, San Jose and Ft Lauderdale. Also happy are cold but coastal Seattle and Manhattan (New York County).

Detroit and Houston are the least happy.In addition to Houston, there are 2 otherunhappy Texas cities, Dallas and Forth Worth. NYC Queens and Brooklyn ar enot too happy either in sharp contrast to very happy Manhattan.

Then few interesting nuances. For instance 3 Texas cities Dallas, Forth Worth (neighbors), and San Antonio score poorly on energy/vitality (first 4 numeric columns) but quite well on happiness (subsequent 6 numeric columns).

At the bottom comparison to deciles from earlier figures (note that comparison colors in the table differ as they are scaled to current table). decile 1 tends to be higher than mean of deciles 2-4

of the largest metropolises some score better or worse than decile averages

Highest and lowest bot energy/vitality and happiness tends to be in small counties, below median population size (see appendix).

ONLINE APPENDIX

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead]

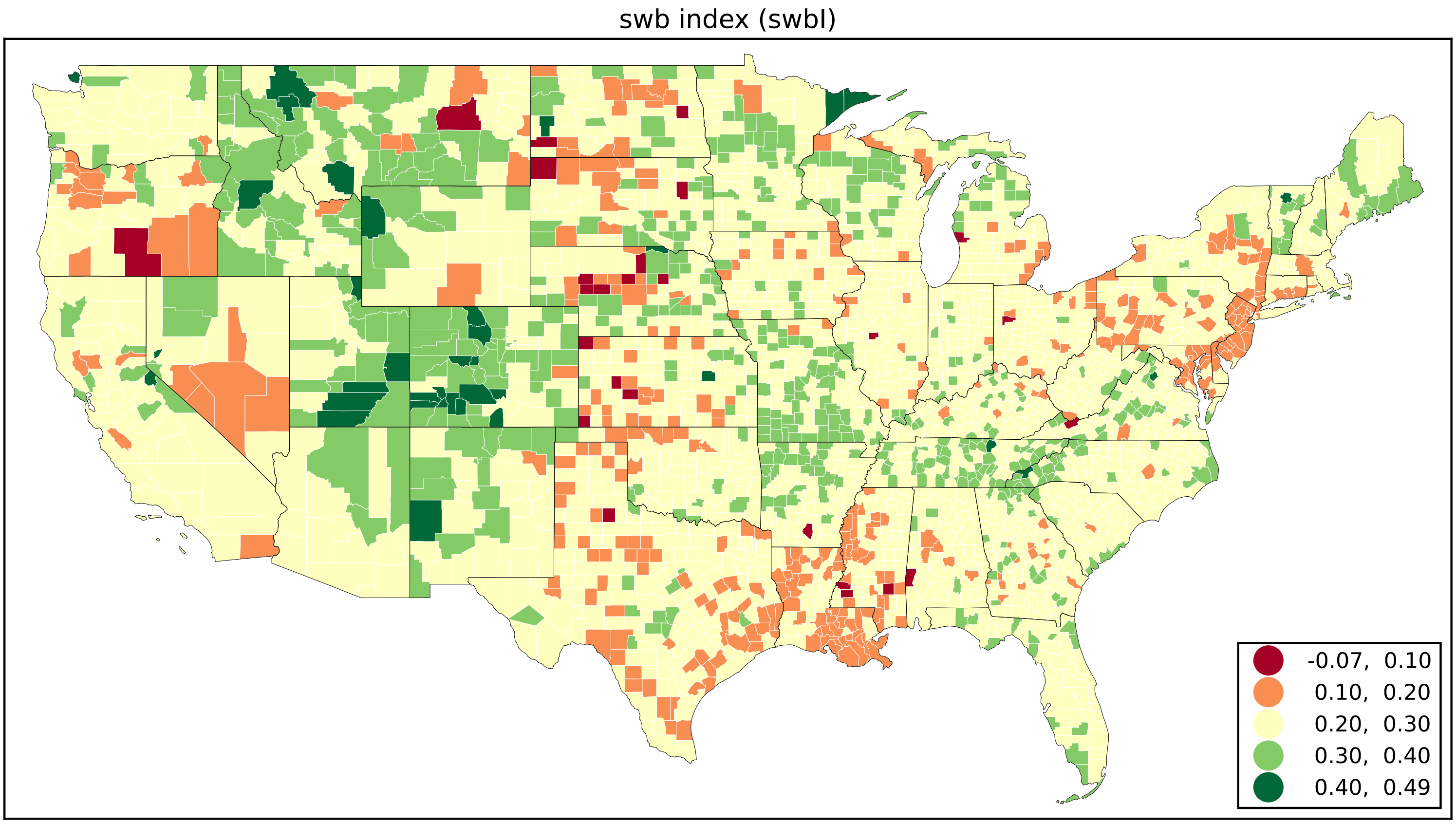

Figure 4: This figure was pulled directly from a web URL.

4.4 Additional results

4.5 Data Description

1) Happiness. The text expresses some level of happiness (`happiness`).

low: "Mondays always feel endless and gray."

med: "Trying to notice at least one small thing that makes me smile today."

high: "We reached our team goal today–feeling genuinely proud and joyful!"

2) Resilience. Capability of withstanding or recovering from difficulties (`resilience`).

low: "Nothing seems to help me move on from this."

med: "It's been a hard season, but we're still finding ways to keep going."

high: "Every setback is just another chance to start stronger than before."

3) Self-esteem. Confidence in one's worth or abilities. (`selfesteem`)

low: "I never feel good enough for anyone."

med: "I might not tick every box, but I know I bring value."

high: "Feeling confident and loving who I'm becoming."

4) Life Satisfaction. Satisfaction with one's life as a whole (`lifesat`).

low: "My living situation is exhausting and unfair."

med: "Things are slowly falling into place–can't complain too much."

high: "Watching the sunset from my porch tonight, feeling truly content."

city	STNAME	CTYNAME	pop.(m)	eneI	posfunct	vitality	energy	swbI	lifesat	happiness	innerp
Los Angeles	California	Los Angeles County	10.1	0.34	0.34	0.26	0.44	0.26	0.12	0.29	
Chicago	Illinois	Cook County	5.2	0.32	0.30	0.21	0.46	0.26	0.11	0.30	
Houston	Texas	Harris County	4.6	0.28	0.26	0.17	0.40	0.17	0.03	0.16	
Phoenix	Arizona	Maricopa County	4.2	0.34	0.33	0.25	0.45	0.27	0.11	0.31	
San Diego	California	San Diego County	3.3	0.36	0.35	0.29	0.45	0.28	0.14	0.34	
Santa Ana	California	Orange County	3.1	0.37	0.36	0.29	0.46	0.29	0.15	0.35	
Miami	Florida	Miami-Dade County	2.7	0.36	0.36	0.28	0.45	0.28	0.14	0.32	
Brooklyn	New York	Kings County	2.6	0.32	0.31	0.23	0.44	0.21	0.06	0.28	
Dallas	Texas	Dallas County	2.6	0.28	0.27	0.18	0.40	0.22	0.04	0.26	
Riverside	California	Riverside County	2.3	0.31	0.30	0.21	0.41	0.24	0.10	0.29	
Queens	New York	Queens County	2.3	0.30	0.28	0.19	0.42	0.18	0.01	0.24	
Seattle	Washington	King County	2.1	0.36	0.35	0.28	0.46	0.28	0.11	0.33	
San Bernardino	California	San Bernardino County	2.1	0.30	0.29	0.20	0.40	0.23	0.08	0.28	
Las Vegas	Nevada	Clark County	2.1	0.37	0.35	0.28	0.47	0.28	0.15	0.34	
Fort Worth	Texas	Tarrant County	2.0	0.27	0.25	0.17	0.39	0.21	0.04	0.25	
San Jose	California	Santa Clara County	1.9	0.35	0.34	0.26	0.46	0.28	0.12	0.33	
San Antonio	Texas	Bexar County	1.9	0.28	0.27	0.17	0.40	0.22	0.05	0.27	
Fort Lauderdale	Florida	Broward County	1.9	0.35	0.34	0.27	0.43	0.26	0.12	0.30	
Detroit	Michigan	Wayne County	1.8	0.25	0.24	0.14	0.38	0.19	-0.01	0.24	
Manhattan	New York	New York County	1.6	0.38	0.37	0.28	0.48	0.28	0.14	0.35	
		decile 1 (<5.1k)		0.35	0.37	0.26	0.43	0.24	0.15	0.24	
		mean of deciles 2-4 (5.1-18.5k)		0.33	0.32	0.23	0.43	0.26	0.10	0.29	
		decile 10 (>204.9k-10077.3k)		0.31	0.30	0.21	0.42	0.24	0.06	0.28	

Figure 5: Energy/vitality and happiness for 20 largest counties. Counties are sorted on population. At the bottom for comparison averages for 1st decile, 2-3rd deciles, and 10th decile.

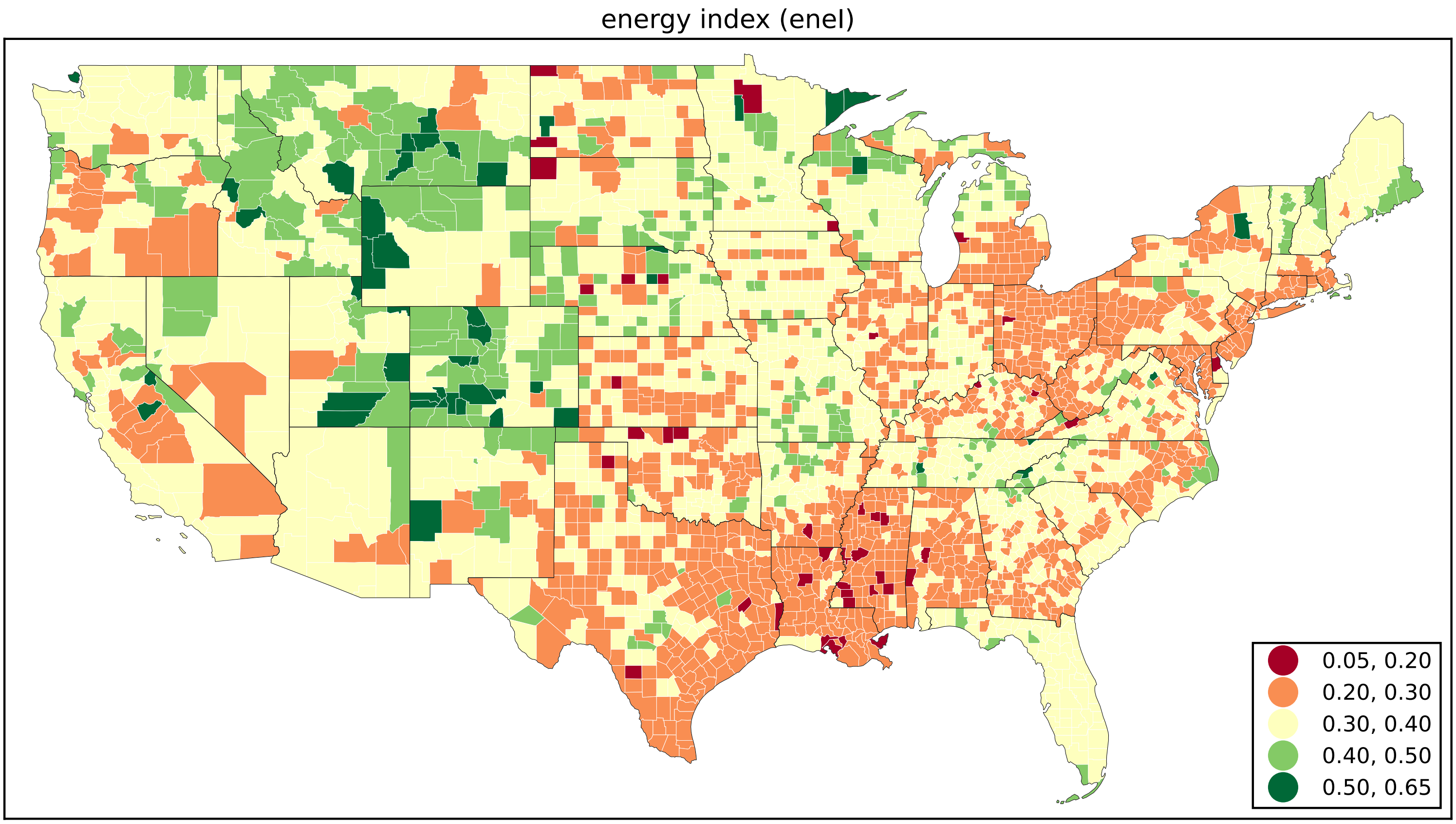

Figure 6: more classes This figure was pulled directly from a web URL.

5) Fear of future. Worry about one's condition in the coming years (`fearfuture`).

low: "I'm sure everything will work out somehow."

med: "Uncertain about how next year will turn out financially."

high: "The future feels like a cliff I'm walking toward in the dark."

6) Vitality. Feelings of strength and activity (`vitality`).

low: "Still recovering from a cold, running on empty."

med: "Tired but determined to finish this week strong."

high: "The energy today is unreal–everything feels possible!"

7) Having energy. Reports of feeling (or not) energetic (`energy`).

low: "Can't seem to get moving this morning."

med: "A little sluggish but getting things done."

high: "So full of energy I could run another mile!"

8) Positive functioning. Feeling capable and effective (`posfunct`).

low: "Everything feels like too much lately."

med: "Getting back into a productive rhythm again."

high: "Handled a dozen tasks today–completely in the zone!"

9) Expressing job satisfaction. Satisfaction with one's present job (`jobsat`).

low: "Work feels meaningless lately."

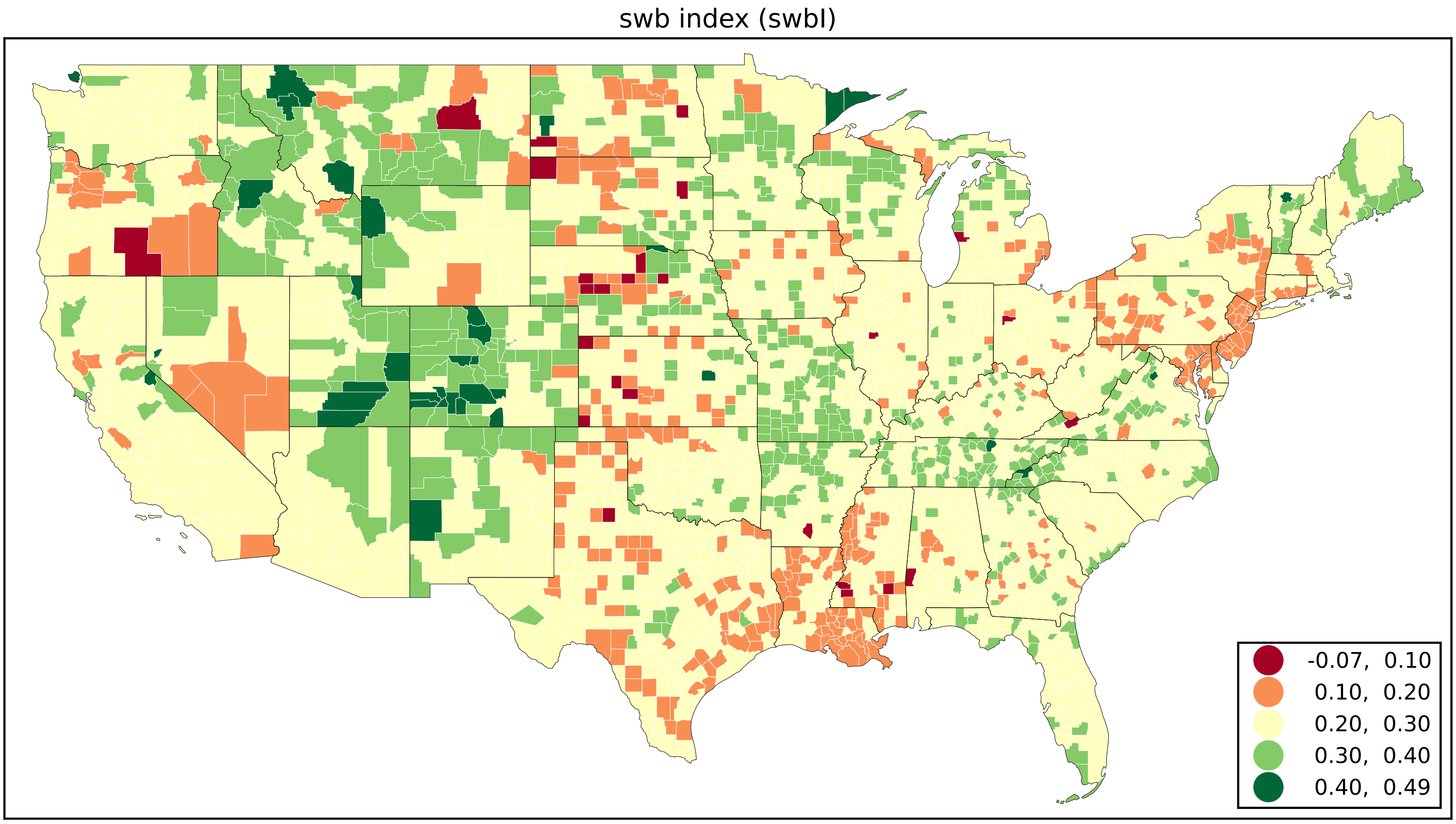

Figure 7: more classes This figure was pulled directly from a web URL.

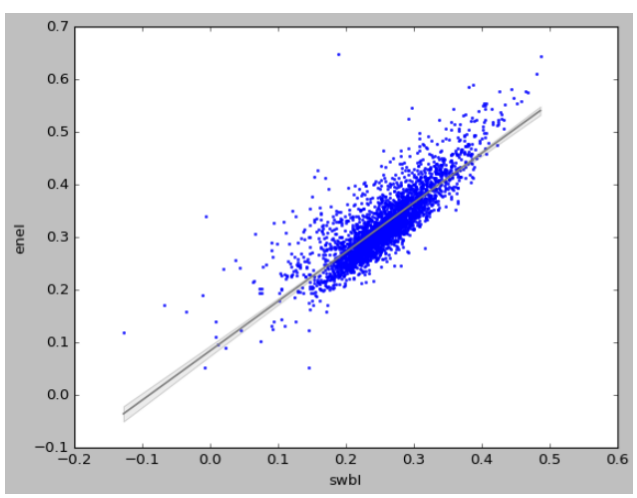

Figure 8: Correlation is 0.83.

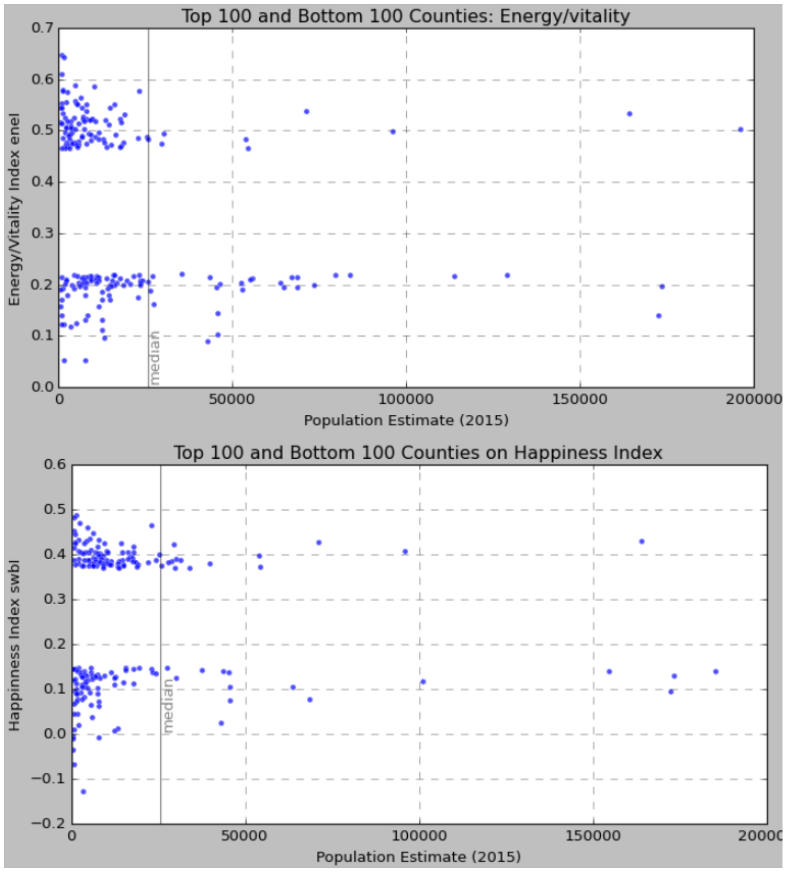

Figure 9: Highest and lowest bot energy/vitality and happiness tends to be in small counties, below median population size.

med: "Busy week but I'm learning new things."

high: "I truly enjoy what I do–it fits me perfectly."

10) Expressing optimism. Optimism about one's condition in the medium run (`optimism`).

low: "I don't expect much to improve soon."

med: "Maybe next month will bring better news."

high: "Things are finally turning around–I can feel it!"

11) Peace with thoughts and feelings. General feeling of calm acceptance (`innerpeace`).

low: "Can't quiet my thoughts enough to rest."

med: "Learning to take things as they come."

high: "Feeling calm and centered after a long day."

12) Purpose in life. Acting with a sense of purpose (`purpose`).

low: "I'm not sure what I'm doing any of this for."

med: "Trying to find direction and meaning in small steps."

high: "Every action today felt aligned with what I believe in."

13) Depression. Expressions of hopelessness or anhedonia (`depression`).

low: "Nothing negative–just a regular day."

high: "Everything feels heavy and empty at once."

14) Anxiety. Feeling nervous or unable to control worry (`anxiety`).

low: "Calm and relaxed morning."

high: "My heart won't stop racing over things I can't fix."

15) Suffering. Expression of physical or mental pain (`suffering`).

high: "The pain won't let me sleep tonight."

16) Feeling pain. Explicit mention of pain or grief (`pain`).

high: "Missing someone so deeply it physically hurts."

17) Expressing altruism. Acting for others' benefit (`altruism`).

low: "Wishing people well but staying uninvolved."

med: "I hope things get better for you soon."

high: "Spent the evening delivering meals to neighbors in need."

18) Loneliness. Expressions of loneliness (`loneliness`).

low: "Remembering old friends with fondness."

med: "Sometimes being alone feels peaceful, sometimes not."

high: "It's like the whole world moved on without me."

19) Quality of relationships. Satisfaction with relationships (`relationships`).

low: "We argue more than we talk lately."

med: "Dinner with family went better than expected."

high: "I'm grateful every day for the people who truly care."

20) Belonging to society. Sense of community belonging (`belonging`).

low: "Feels like I don't fit anywhere anymore."

high: "Proud of my community for showing up today!"

21) Expressing gratitude. Feeling and expressing gratitude (`gratitude`).

high: "So thankful for the help and kindness I received today."

22) Expressing trust. Trust toward people or institutions (`trust`).

low: "Hard to rely on anyone lately."

med: "I'll give them another chance."

high: "I know they'll do the right thing."

23) Feeling trusted. Reporting that one is trusted by others (`trusted`).

low: "They still don't believe in me."

high: "Being asked for advice feels good–it means they trust me."

24) Balance in life. Maintaining equilibrium across life domains (`balance`).

low: "Work completely took over my week."

med: "Still learning to leave the laptop closed after 6."

high: "Spent the morning running and the evening with friends–perfect balance."

25) Mastery (ability/capability). Feeling competent (`mastery`).

low: "Everything I try seems to go wrong."

med: "Finally getting the hang of this project."

high: "Solved a tricky problem today–felt amazing!"

26) Perceiving discrimination. Feeling discriminated against (`discrim`).

med: "People keep mispronouncing my name–it gets tiring."

high: "I was turned away just because of where I'm from."

27) Feeling loved by God. Feeling loved by a higher power (`lovedgod`).

high: "I feel surrounded by divine care and peace."

28) Belief in God. Belief in a divine entity (`believegod`).

med: "Faith gives me comfort even when I doubt."

high: "Trusting completely in God's plan today."

29) Religious criticism. Feeling criticized for one's beliefs (`relcrit`).

high: "They mocked my faith again, but I won't hide it."

30) Spiritual punishment. Feeling punished by a spiritual force (`spiritpun`).

high: "Maybe this hardship is a test from above."

31) Feeling religious comfort. Finding comfort in spirituality.

high: "Prayer always brings me back to calm and strength."

32) Financial/material worry. Concern about finances (`finworry`).

low: "Joking about how expensive coffee is today."

high: "Not sure how I'll cover rent this month."

33) Life after death belief. Belief in life after death (`afterlife`).

high: "I believe our stories continue beyond this world."

34) Volunteering. Giving time for others (`volunteer`).

high: "Spent the weekend mentoring kids at the community center."

35) Charitable giving/helping. Donating or assisting others (`charity`).

high: "Joined a fundraiser–every bit helps someone in need."

36) Seeking forgiveness. Willingness to forgive (`forgive`).

high: "I've decided to let go and forgive–it feels lighter."

37) Feeling having a political voice. Feeling empowered to influence politics (`polvoice`).

med: "Signed a petition today–not sure it matters, but I tried."

high: "Spoke at a town meeting–change starts small but real."

38) Expressing government approval. Expressing satisfaction with governance (`govapprove`).

high: "The new policy finally addresses what we've been asking for."

39) Having hope. Hope about the future (`hope`).

low: "Hard to see a way forward."

med: "Things might improve if we stay patient."

high: "Tomorrow will be brighter–I truly believe it."

40) Promoting good. Acting to promote the common good (`goodpromo`).

high: "Planted trees with neighbors today–small steps for a greener town."

41) Expressing delayed gratification. Willingness to wait for future reward (`delaygrat`).

low: "Can't resist buying it now."

med: "Saving a bit each week for the trip."

high: "Working hard today for the long-term goal ahead."

42) PTSD. Bothered by past traumatic experiences (`ptsd`).

med: "Still have flashbacks sometimes."

high: "Therapy is helping me process what happened years ago."

43) Smoking-related health issues. Mentions of smoking and health (`smokehealth`).

high: "Trying again to quit–breathing feels tougher lately."

44) Drinking-related health issues. Mentions of alcohol use or effects (`drinkhealth`).

med: "Had a few drinks too many, need to slow down."

high: "Realizing drinking is hurting my health–it's time to change."

45) Health limitations. Health problems restricting daily activity (`healthlim`).

low: "Catching a small cold, nothing serious."

med: "Back pain keeps me from sitting too long."

high: "Severe migraine kept me in bed all day."

46) Expressing empathy. Showing compassion for others (`empathy`).

low: "Everyone deals with their own stuff–don't ask me to care."

med: "That story was moving; hope they recover soon."

high: "My heart breaks for the families affected–I wish I could help."

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